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BIO TEST - 2

Date _____
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Section - A

Question - 1

(c) Artificially acquired passive Immunity

Question - 2

(b) Metaphase - 1

Question - 3

Stationary phase . . . ?

Question - 4

ZIFT - Zygote Intra-fallopian Transfer !!

Question - 5

Low dissolved Oxygen content in the body due to the presence of unwanted organic compounds.

Question - 6

(b) Cleistogamy

Question - 7

Blastula stage (Blastocyst)

Question - 8

Opium Somniferous

Question - 9

$$p^2 + q^2 + 2pq = 1$$

Question - 10

bactulum -

Section - B

Question - 11

It is a form of asexual reproduction in which seeds are produced without fertilisation.

Question - 12

Divergent

Convergent

- Same basic structure & origin
- Diverge to ~~per~~ do different things

- Different structure & origin
- Adapt to do similar things

Question - 13

- It acts as a plant growth promoter.

Question - 14

Test cross

Back cross

- A cross between an F_1 gen and its parent individual in the F_1 generation.
- A cross between an individual showing a dominant trait and its recessive parent.

Question - 15

- Female foeticide: Strict laws help prevent the misuse of ~~press~~ techniques that help determine the baby's fetus's gender.
- Mother's health & safety: To ensure that abortions are only performed by licensed doctors and only for legal reasons.

Question - 16

Normal

Cancerous

- Exhibit Contact Inhibition when they come in contact with the neighboring cells.
- Single layer
- Do not exhibit contact inhibition
- Tumor

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Question - 17

- Leaf modification: Leaves are found with spine or scales to minimise surface area for water loss.

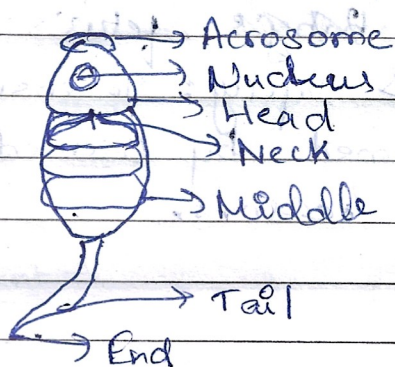
Section - C

Question - 18

LH - Acts on Leydig cells of the testes and stimulates testosterone, which in turn stimulates the process of spermatogenesis.

FSH - Acts on Sertoli cells which stimulates inhibin, which is required for spermatogenesis.

Question - 19



Question - 20

Male birds are homozygous (ZZ) and female birds are heterozygous (ZW).

The sex of the offspring is determined by the female gamete.

In humans, male is heterozygous (XY) and female is homozygous (XX Chromosomes). The sex of the offspring is determined by the father's gamete.

Question-21

Mammals such as humans, bats, whales, cheetahs, etc. have the same skeletal elements like radius, ulna, carpal, metacarpal and phalanges. Though they show a major functional divergence, they have a similar structural descent. This shows that homologous organs are evidences for evolution.

Question-22

Oligospermia is the absence of sperm in the male. ~~ICSI~~ - Intracytoplasmic Sperm Injection.

~~IVF~~ - ~~Intra-uterine Insemination~~ is a technique where a single, motile sperm is isolated and injected directly into the cytoplasm of a mature ovum, in artificial conditions.

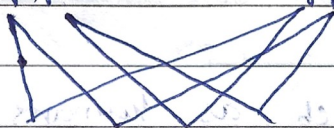
Once fertilisation occurs, it is transferred into the female's fallopian tube. This removes the necessity for a high sperm count.

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Question-2

In incomplete dominance, neither allele is completely dominant over the other. In the heterozygous state, an intermediate phenotype is expressed.

Parents: $RR \times rr$ (RR - red flowers, rr - white)



Rr Rr - Rr Rr

F_1 gen: All Rr - Pink flowers

$Rr \times Rr$

$\times$	R	r
R	RR (red)	Rr (pink)
r	Rr (pink)	rr (white)

Phenotypic ratio: 1 red : 2 pink : 1 white

Genotypic ratio: 1 RR : 2 Rr : 1 rr
(1 : 2 : 1)

Section - DQuestion - 25Option - B

(a) • Monochromatic unit : Composed of 3 components

1) Nitrogenous base: Purines & Pyrimidines

2) Pentose sugar: deoxyribose

3) Phosphate :

• Linkage

1) N-glycosidic bond : Links the nitrogenous base to the C₁ of the pentose sugar, forming deoxyribonucleoside.

2) Phosphodiester bond : 3'-5' connection, forming the sugar-phosphate backbone of the DNA.

(b) Central Dogma of Molecular Biology describes the unidirectional flow of genetic information

DNA $\xrightarrow{\text{Transcription}}$ mRNA $\xrightarrow{\text{Translation}}$ Protein

• Enzyme responsible for reversing the flow of information is Reverse Transcriptase.

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Question - 26

Option - A

(a) Double Fertilisation:

- Syngamy - One of the 2 haploid male gametes delivered by the pollen tube fuses with the haploid egg cell. This forms a diploid zygote which develops into an embryo.
- Triple fusion: A second haploid male gamete goes to the cell and fuses with 2 polar nuclei. It forms a triploid Primary Endosperm Nucleus which develops in nutrient-rich endosperm.

(b) The Synergids etc

- The synergids degenerate after fertilisation
- The antipodal cells serve as nutrition and then degenerate ^{right after} ~~before~~ fertilisation.

Question - 27

Option - B

1) Menstrual phase :- 1-5 days

- Unfertilised egg degenerates
- Low levels of LH & FSH
- Oestrogen & Progesterone levels drop rapidly, which causes the breakdown of the endometrium

2) Follicular phase :- 6-13th day

- Primary follicles develop ~~through~~ into Graafian follicles.
- FSH rises, LH increases mid-cycle.

- Increasing amount of Oestrogen, which help repair the endometrial lining

3) Ovulatory phase: • Day 14

- Graafian follicle ruptures, releasing a secondary oocyte into the uterus.

- Surge of LH at its peak, FSH also peaks

- Peak concentration of Oestrogen causing positive feedback

4) Luteal phase: • Days 15-28

- The ruptured follicles becomes Corpus Luteum. In the absence of fertilisation, it ~~becomes~~ degenerates to form Corpus Albicans.

- LH and FSH levels decline & remain low due to negative feedback

- High levels of Progesterone which maintains the endometrial lining for potential implantation.

Question - 23

- It uses tungsten electrodes for analyses of compounds in the air (NH_3 , H_2O , CO_2 , etc.).
- It proves that monomers could form ~~spont~~ spontaneously under primitive conditions.